🔧 Engine Teardown & Rebuild: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 8 · Level 3 · goes with Engine project + Engine Assembly 1 + Ring End Gap Handout

Name: _____

Question: Is your engine's piston and valve condition within spec, and does the rebuilt engine run?

Materials

engine, tool set, flywheel puller, feeler gauges, torque wrench, lapping kit, parts tray, manual

⚠️ Safety

- Safety glasses and coveralls
- Plug wire disconnected; fluids drained with adult supervision
- Springs: eye protection, tools with adult supervision
- Start engine outdoors with an adult

Prediction

Predict which part will be most worn.

Answers vary; often rings or exhaust valve.

Procedure

1. Tear down in order with photos and labeled trays.
2. Measure ring end gap and side clearance.
3. Lap valves; set clearance.
4. Reassemble with timing aligned and bolts torqued.
5. Test fuel, spark, compression; run with adult.

Data

Check	Spec	Measured	Action
Top ring end gap	**0.007-0.017**	**0.025**	**replace rings**
Ring side clearance	**0.002-0.004**	**0.003**	**OK**
Intake valve clearance	**0.005-0.007**	**0.006**	**OK**
Exhaust valve clearance	**0.007-0.009**	**0.004**	**grind stem**
Head bolt torque	**140 in-lb**	**140 in-lb**	**done**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: what was wrong with your engine and did the rebuild fix it? Evidence: your measurements and test. Reasoning.
Example: worn rings (large end gap) caused low compression; new rings raised compression and it started.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*